

ANDRES BUELVAS

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SUMMARY

Computer engineer with a strong background in web development, driven by a passion for creating robust, scalable, and reliable systems and software. Proficient in leading, designing, implementing and executing components and functionalities with specialization in fintech and strong interest in cybersecurity and artificial intelligence. Dedicated to delivering top-quality software that makes an immediate impact on customers through adherence to high standards of quality. I am a quick learner, dedicated, and thrive in team environments. Continuously seeking opportunities to leverage emerging technologies to enhance my technical expertise and leadership.

EXPERIENCE

Juristat

May 2025 – Sept 2025

Frontend Engineer / TypeScript, Remix, PostgreSQL, TailwindCSS

- **Optimized high-volume data queries** using synchronization and caching strategies.
- I used **Remix** to develop data visualizations, including charts, tables, and reports.
- I created a new inline filter using **Shadcn/UI** and **Tailwind**, designed to handle multiple data types and queries, which enabled its reuse across various application sections.

SmartHop

Apr 2019 – Jan 2025

Software Engineer IC5 / Engineer Manager / Node.js, React, MongoDB, Redis, Stripe

Feb – Jan 2025

- I led a team of up to 4 developers as the **junior technical leader**, responsible for designing, implementing, and maintaining a **Fintech** product that included modules for *invoices, payrolls, wallets, fuel, and billing*.
- Overseeing, designing, and implementing the invoicing module for accurately recording service delivery revenues.
- Creation of *SmartPay*, a secure **funding** mechanism that utilizes **ACH accounts** for user payments.
- Supervision, design, and execution of integration with *UNIT's API* for the creation of **bank accounts**, generation of **debit cards**, and facilitating transfers between internal and **ACH accounts**.
- Implementation of **2-Factor** mechanism as a security requirement for integration with *UNIT*.
- Implementation of *Onramp* integration for generating **fuel cards** and visualizing transactions.
- Use **GraphQL** to visually map gas stations within a specified distance radius and easily locate the nearest ones using **Google Maps** and **Trimble Maps API**.
- Management, deployment, and maintenance of the *Payroll module (V2)*, featuring on-demand generation of income and expense records, along with *PDF* report display capabilities.
- Creation of a **ledger** system that accurately records all income and expenses in accordance with **accounting** principles.

Senior Full Stack Engineer / JavaScript, Node.js, React, MongoDB, Redis, Redux, Stripe

Aug 2020 – Feb 2021

- I have been appointed as the exclusive developer responsible for overseeing all products pertaining to *e-commerce, fuel, and insurance* within the company.
- Integration with the *EFS/WEX API* using **SOAP** for fuel card generation and visualization of transaction activities. The integration architecture followed the **OOP model**.
- Integration with **PLAID**, a powerful library that enables secure client authentication (*OAuth*) within banking institutions, ensuring the safe handling of sensitive banking information.
- Responsible for designing, implementing, and maintaining the *Payroll module (V1)* infrastructure.

Full Stack Engineer / JavaScript, Python, Node.js, React, MongoDB, Redis, Redux, Stripe

Apr 2019 – Aug 2020

- Developing a dynamic dashboard using **Chart.js** and **React** to visually represent user profit through interactive charts.
- Integration of the **Stripe API** for payment plan subscriptions and secure management of customer payment methods.
- Create a password recovery feature that enables users to reset their password securely.
- Utilizing **geolocation queries** and **set theory** to efficiently retrieve resources in **Redis**.
- Developed email implementation and sending functionality using **Mandrill** and **Mailchimp**.

CIDITEL-VE / Latinux Inc

Jul 2018 – Apr 2019

Backend Engineer / Python, Web2py, Bootstrap

- Developed **MECE** system, introducing an alternative method responsible for enabling exam takers to gain admission to the Simon Bolivar University as students.
- Developed an online academic exam system that includes a time limit for each category (*numerical and verbal ability*).
- Developed the teacher's capability to initiate the online academic exam for all students once they are ready, and to promptly receive feedback on the results upon completion of the exams.

Simon Bolivar University

Apr 2018 – July 2018

Support Assistant / Python, Web2py, Bootstrap

- Developed **SisPIO** system that empowered numerous students to enhance their skills for the university exams.
- Project utilized for administrative purposes, including control, student tracking and *PIO* program activities.

PROJECTS

Document Parser | *Python, FastAPI, Langchain, Supabase, AWS, Mistral AI, Langchain, OpenAI*

- Developed a **FastAPI** system capable of classifying, processing, and storing document types from images or PDF files.
- Extracted information using libraries and tools like **Tesseract OCR** for reading PDFs and images.
- Designed and implemented a state machine to provide full traceability and control over the document processing lifecycle, including analysis, classification, segmentation, extraction, validation, and final storage.
- Utilized Supabase for cloud database management and **S3** buckets for file storage.
- Integrated **LangChain**, **OpenAI** and **Mistral AI** with prompt engineering to enhance the precision of document analysis, classification, and data extraction.

Account Receivable (AR) System | *TypeScript, Nest.js, Docker, RabbitMQ*

- Developed a middleware system for automated communication between *FSG* and *Service Channel*, handling work order lifecycle, notes, invoicing, and notifications.
- Designed a **NestJS** application with a Domain-Driven Design (DDD) and a 4-layer architecture for robust component, service, and data access separation.
- Utilized **RabbitMQ** to implement an asynchronous queue model for task management.
- Deployed a webhook mechanism to facilitate communication between providers and the AR system.
- Integrated state machines with the queuing system to provide full traceability, error handling, and logging for key events like work orders and invoices, ensuring a flexible and monitored workflow.

Photo Post | *Node.js, Express, Jest, SQLite, Redis, Kafka, Docker, AWS*

[Source Code](#)

- Microservice-based platform built with **Node.js** and **Express**, ensuring seamless RESTful API communication.
- I used **Docker** for containerization with persistent volumes for images and databases, Sequelize with **SQLite** for data management, **Redis** for caching, and **Kafka** for asynchronous messaging between microservices.
- Designed for scalability and modularity, each service maintains a single responsibility while robust authentication (with passport and bcrypt) and testing (via **Jest**) guarantee secure and reliable performance.

CQRS Blog Application | *Node.js, React, Tailwind, Schaden/UI, MongoDB, MySQL, RabbitMQ*

[Source Code](#)

- Development of a Blog API utilizing a microservices-based architecture and the **CQRS** design pattern.
- Implementation of an API gateway for request redirection, along with a command service for write operations on a **MySQL** database and a query service for read operations using **MongoDB**.
- Asynchronous communication between services using **RabbitMQ**.

ClueGarden | *JavaScript, Node.js, Firebase, Heroku, Stripe*

- Developing, analyzing, and implementing the infrastructure for an *e-commerce* system platform that utilizes **Stripe** as a payment gateway to enable users to subscribe to various plans and services.
- Develop **APIs** and seamlessly integrate them with the application's user interface. Implemented **Firebase** communication as a **Backend as a Service (BaaS)** to leverage **Cloud Firestore** as the cloud database solution.

Portafolio 2024 | *JavaScript, Next.js, Tailwind, Schaden/UI, GitHub Actions*

[Source Code](#)

- I have developed a **Next.js** project to demonstrate my diverse range of academic, professional, and technical skills.
- The view was developed using **Tailwind** and **Schaden/UI** to incorporate a dark-light mode, accordion cards for displaying projects, and enhancing the overall website layout.

GPI System | *React, Django, MongoDB, Swagger, Material UI*

- University project that implemented academic exam management by monitoring students' knowledge and results.
- Implemented personalized exam complexity adjustments based on individual student levels to strengthen areas of weakness and reinforce previously acquired knowledge.
- Developed a **real-time application** for conducting academic exams, which showcases data through interactive charts.

EDUCATION

Simon Bolivar University

2013 - 2019

Bachelor in Technology (BTech) in Computer Engineering

Caracas, Venezuela

Pontificia Universidad Javeriana

2025 - 2025

Diploma in Software Architecture and Cloud Computing

Cali, Colombia

COURSEWORK

Modern Software Architecture: DDD, Events, Microservices	Apr 2025
Master in Microservices with JavaScript and Node.js	Mar 2025
AWS Cloud Technical Essentials	Jan 2025
EF SET Certificate Writing (C2), Reading (C1), Listening (B2), Speaking (B2)	Jan 2025
Programming in Python	Apr 2024
Clean Code	Mar 2024
Version Control	Mar 2024
Introduction to Front-End Development	Feb 2024
Introduction to Generative AI	Feb 2024
Programming with JavaScript	Feb 2024

SKILLS

Languages: JavaScript, TypeScript, Python, MATLAB.

Backend: Node.js, Express.js, Nest.js, Kafka, RabbitMQ, Web2py.

Frontend: React, Remix, Redux, Vite, Tailwind, Next.js, HTML5, CSS, Material UI, Bootstrap, Shadcn/UI.

AI: MCP, Langchain, Mistral AI, OpenAI.

Testing: Jest, Puppeteer.

Containers, Cloud & Databases: MongoDB, MySQL, PostgreSQL, Redis, Firebase, GitHub Actions, Bitbucket Pipelines, Google Cloud, AWS, S3, Supabase, DynamoDB, API Gateway, Lambda Functions, EC2, ECR, ECS, Fargate, Cloudwatch, Docker.

Developer Tools: Git, Postman, Insomnia, Jira, Bitbucket, VSCode, Cursor, WindSurf, GitHub, Swagger

Concepts: Relational and Non-Relational Database, Object-Oriented Programming (OOP), Cache Memory, Big O Notation, Agile Methodology, Cloud Computing, Real-Time Applications, Asynchronism, Synchronism, Threads, Multi-Threads, ServerLess, Infrastructure as Service (IaaS), Platform as a Service (PaaS), Software as Service (SaaS), Continuous Integration & Continuous Delivery (CI/CD), Monolithic & Microservices Architecture, Auto Scaling and Load Balancing, API REST and Gateway, GraphQL, API Integrations, Event-Driven Architecture, MVC Architecture, Hexagonal Architecture, Design Patterns, CQRS, IA Agents, Model Context Protocol, Prompting Engineering, Artificial Intelligence Agents.